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Three brain collections for comparative neuroanatomy and neuroimaging

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In the context of increasing extinction rates and the potential loss of essential evolutionary biological and anthropological information, it is an important task to support efforts to prepare, preserve, and curate collections of histological brain sections; to disseminate information on such collections in the neuroscience community; and to make the collections publicly available. This review emphasizes the importance of complete, serially sectioned human brains of different ontogenetic stages as well as those of adult and old human individuals for neurobiological and medical research. Such histological sections enable microstructural analyses and anatomical evaluations of functional and structural neuroimaging data, for example, based on magnetic resonance imaging. Here, this review provides the first detailed and updated account of the content of the Stephan, Zilles, and Zilles-Amunts collections, which consist of serially sectioned and cell body- and myelin-stained histological preparations. Finally, this review will give an overview of past and recent research using these collections to increase our understanding of the detailed patterns of divergent brain evolution in primates as well as of the structural organization of the human brain.

Keywords: brain collections; comparative neuroanatomy; neuroimaging; brain mapping; cytoarchitectonics

Introduction

Comparative brain data spanning beyond humans, chimpanzees, and macaques are imperative for modeling the intricacies of complex evolutionary patterns underlying divergent brain evolution in primates. Detailed evolutionary models are essential when inferring brain–behavior interactions, selective pressures underlying structural changes, and genetic underpinnings of brain development. This goal can be considerably supported if brains of other groups of vertebrates are available, particularly if the selection of species covers nearly completely a group like insectivores or bats, because these species show a wealth of different adaptations, evolutionary changes, nutritional habits, and ecological environments. For example, in the last 10 years, information has become available with regard to genotype differences between primates. Information on

species-specific variation in primate brain structure evolution has, however, lagged behind the exponential advances in genotype research, thus limiting the information that can be deduced from genotype–phenotype comparisons. It is clear that the more information is available on detailed evolutionary patterns of brain structure changes, the more useful the existing genotype data become in inferring possible candidate genes for particular brain structures. Cross-species phenotype information thus functions as a direct bottleneck (or catalyst) for further advances in genotype–phenotype comparisons and the medical and biological information they may provide.

The general argument is straightforward and applies beyond genotype–phenotype comparisons: the more species/individuals for which data are available, the more specific the inferred evolutionary patterns and species-specific differences are and the

more accurate the evolutionary information is that can be deduced. Efforts to collect and curate brain collections and data on many different species play a fundamental role in extracting the evolutionary information hidden in extant variation. Moreover, without support for such efforts, essential medical, neuroanatomical, evolutionary biological, and anthropological information may be lost forever as species extinction is increasing at alarming rates. Here, we provide an updated account on the content of the Stephan, Zilles, and Zilles-Amunts collections and how recent research has used these data to increase our understanding of human brain organization and the detailed patterns of divergent brain evolution.

Short catalogs of the collections

The Stephan collection consists of serially sectioned—in most cases whole—brains of insectivores, bats, and primates (see catalog in Supporting information). The sections are cell body-stained (Nissl-stained using cresylviolet), cell body-stained using the Merker method,¹ and myelin-stained (various techniques). The collection of insectivores contains 227 brains of 51 different species, the bat collection has 761 brains of 268 species, and the primate collection has 200 brains of 51 different species; additionally, there are 112 brains of three different species of scandentia. Furthermore, 68 brains from 20 different mammalian species are also part of this collection. Since several brains per species are available in many cases, sections in coronal, horizontal, or sagittal planes are prepared. The brains and sections were collected and processed between approximately 1950 and 1980 by Heinz Stephan and others. The collection was donated by Heinz Stephan to Karl Zilles.

Numerous original papers have been published based on the Stephan collection. The publications are focused on the evolution of brain size,^{2–6} gyrification,^{7,8} telencephalon,⁹ the allocortex and its different regions (limbic system,¹⁰ amygdala,^{11,12} septum and septal nuclei,^{13–17} rhinencephalic structures,^{18–20} and the hippocampus^{21,22}), neocortex,^{23–28} striatum,²⁹ and brain stem^{30–34} in primates and insectivores, as well as scandentia.³⁵ Furthermore, monumental monographs on brains of primates,³⁶ insectivores,³⁷ and bats^{38–40} have been published. The quality of the histological material is demonstrated by various examples in Figures 1–3.

Table S1 gives a survey of the content and main data of the Stephan collection.

The Zilles collection consists of serial sections of brains of various species including reptiles (six brains from four different species), marsupials (four brains from three different species), insectivores (15 brains from four different species), rodents (10 brains from four different species), and nonhuman primates (84 brains from 28 different species; see catalog in Supporting information). The sections are cell body-stained (Nissl-stained using cresylviolet), cell body-stained using the Merker method,¹ and myelin-stained (various techniques). In some cases, several brains per species are available that are sectioned in coronal, horizontal, or sagittal planes. The brains were collected by Heinz Stephan, Katarina Semendeferi, and Karl Zilles. The sections have been prepared from approximately 1977 to the present by Karl Zilles with technical support from Ursula Blohm and others. This collection has been used for various published comparative neuroanatomical studies.^{41–60} The quality of the histological material is demonstrated by various examples in Figures 1–3. Table S2 gives a survey of the content and main data of the Zilles collection.

The Zilles-Amunts collection consists of serial sections of complete, undissected human embryos and fetal as well as adult brains (see catalog in Supporting information). Forty-two adult brains (age between 25 and 95 years) were studied additionally with a 3D-sequence using magnetic resonance imaging (MRI) before they were processed for histology. The sections are cell body-¹ and myelin-stained using various techniques, for example, Merker cell body staining, Heidenhain–Woelcke myelin staining, or Gallyas myelin staining. The collection also contains 22 prenatal, 49 neonatal/postnatal (up to 13 years), and 38 adult human brains (age between 20 and 86 years). The brains are sectioned in coronal, horizontal, or sagittal planes. The sections of this collection have been prepared from approximately 1977 to the present by Karl Zilles and Katrin Amunts with technical assistance from Ursula Blohm, Ferda Kocaer, and others. This collection is still growing, and has been used for numerous cytoarchitectonic, probabilistic mapping, and comparative neuroanatomical studies.^{42–58,61–69} Brains that underwent MRI before histological

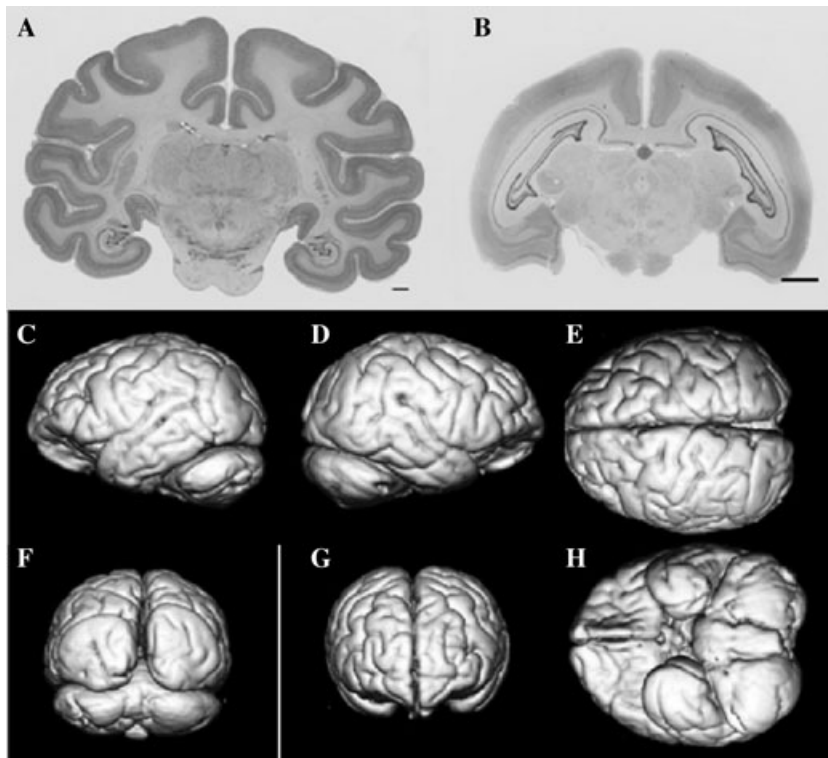


Figure 1. Histological, cell body–stained coronal sections through the brains of *Hylobates lar* (Gibbon) (A) and *Galago senegalensis* (Senegal Bushbaby) (B), both from the Stephan collection. C–H show 3D-reconstructions of the brain of *Pan paniscus* (Bonobo) based on MRI studies of the brain before histological processing (source: Zilles collection). Scale bars in A and B = 2 mm.

processing are used for comparison with functional MRI studies or positron emission tomography (PET) studies. Figure 1 gives an example of the MRI-based reconstruction of a bonobo before histological processing. Table S3 gives a survey of the content and main data of the Zilles-Amunts collection.

Neuroimaging and brain collections: approaches to study structure–function relationships

Human brain atlases gain more and more importance in modern neuroimaging techniques including MRI.⁶⁵ The spatial resolution of routine anatomical MRI enables a localization of the detected brain activities at a macroscopical level, for example, gyri, sulci, and subcortical nuclei (e.g., basal ganglia, thalamus), but does not show the underlying microstructure and cellular architecture. The cytoarchitecture—that is, laminar distribution of cell bodies of the cerebral cortex—is character-

istic for a cortical area and reveals its structural segregation. Cytoarchitectonic research in the human brain became famous with Brodmann's map.⁷⁰ Brodmann's idea, that every cortical area subserves a certain function in the human brain, is still a working hypothesis in many functional imaging studies. Complete serial sectioning of human postmortem brains represents, therefore, an important prerequisite for the three-dimensional reconstruction of the brain's microstructure. The comparison with data from (functional) imaging techniques in a spatial reference frame directly integrates functional and architectonical information.^{67,69} So far, 28 human brains from the Zilles-Amunts collection have been three dimensionally reconstructed using MRI data obtained prior to histological processing in combination with advanced tools of image enhancement, artifact correction, and registration.^{61,67,71} Numerous studies employing the cytoarchitectonic maps based on analyses of the Zilles-Amunts collection can be found in the literature, and the maps are part of several software tools⁷² and brain atlases

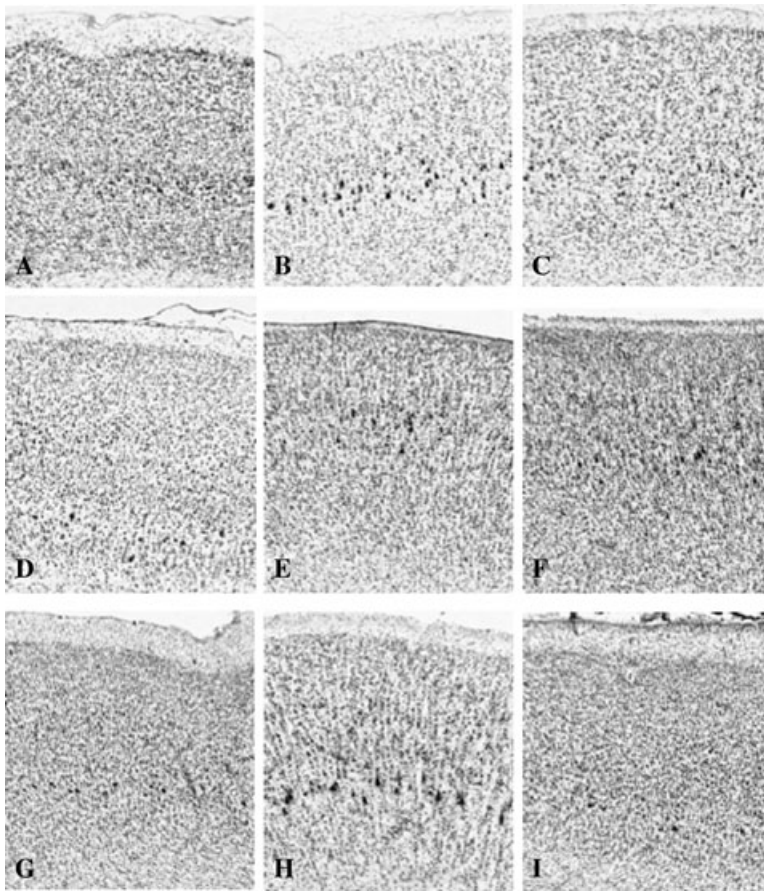


Figure 2. Cell body-stained histological sections through the primary motor cortex (Brodmann area 4) of *Microcebus murinus* (A), *Lemur catta* (B), *Callithrix jacchus* (C), *Aotes trivirgatus* (D), *Macaca mulatta* (E), *Hylobates lar* (F), *Pongo pygmaeus* (G), *Gorilla gorilla* (H), and *Pan paniscus* (I). The typical giant cells of Betz are visible in layer V of all these primate species. Sources: Stephan and Zilles collections.

(Fig. 4). For the cytoarchitectonic mapping work based on brains of this collection, use the following websites:

<http://www.fz-juelich.de/inm/inm-1/index.php?index=307>, <http://www.fz-juelich.de/inm/inm-2/>.

For information about all three collections, contact Karl Zilles or Katrin Amunts, or use the following links:

http://www.fz-juelich.de/inm/inm-2/Brain_Collections, http://www.fz-juelich.de/inm/inm-1/Brain_Collections.

Gyrification in adult and developing primate brains

The three brain collections have enabled a series of studies on the regional and quantitative aspects of

brain folding because a diversity of brains of different primate species and of human brains during different developmental stages is available. For these types of studies, the gyrification index method (GI) was developed.⁷³ Using this method, it could be shown that the gyrification is particularly intensive in the multimodal association regions of the human cortex. This finding was also demonstrated in the cortex of the orangutan⁶⁸ and in the developing human brain.⁴² Moreover, it was shown that two-thirds of the human cortical surface are hidden in the sulci, and the process of gyrification is regionally of different intensity and speed during pre- and postnatal development. Based on the Stephan and Zilles collections, quantitative measures of gyrification in primates are published covering a

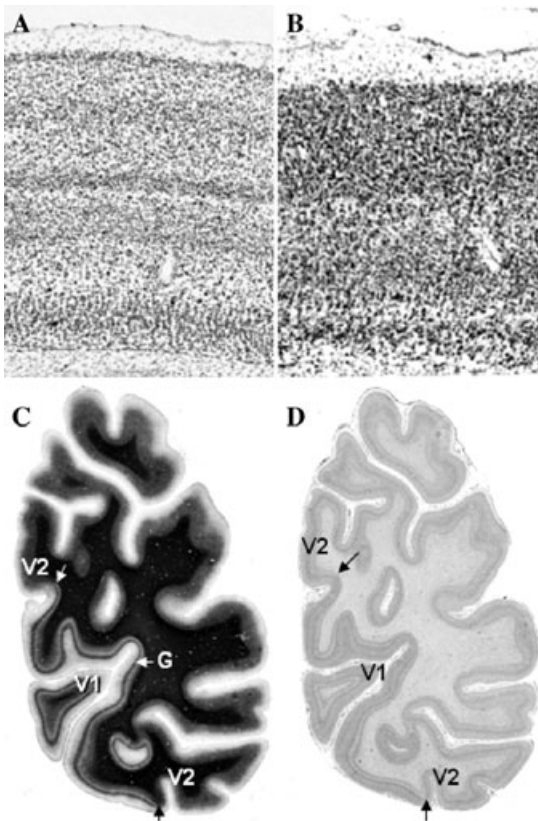


Figure 3. Cell body–stained histological sections through the primary visual cortex (Brodmann area 17; V1) of *Tarsius bancanus* (A) and *Homo sapiens sapiens* (B); C and D show myelin-stained and cell body–stained; D, immediately neighboring coronal sections through a human brain. The primary and secondary visual areas are visible. G: Gennari stripe in V1. Sources: Stephan, Zilles, and Zilles-Amunts collections.

broad range of species.^{8,44,46} Finally, the mechanical folding hypothesis was tested by using the GI as a quantitative measure.⁴⁵

Gray level index

The gray level index (GLI) method was developed to provide a quantitative estimate of the proportion between the volume occupied by cell bodies and that assigned to the neuropil in large samples of brain sections.^{69,74–76} The GLI is of general interest because the neuropil compartment contains the dendrites, axons, and synapses, that is, the structural basis of connectivity. This allows the measurement of a parameter, which may be functionally even more important than the number of cell bodies per volume unit. Since the volumes of glial pro-

cesses and blood vessels, which are also included in the automated measurements of the GLI, are relatively constant throughout cortical layers and regions, a high GLI indicates densely packed cell bodies but a relatively small volume proportion available for connections between nerve cells, for example, primary sensory cortices show a high GLI, whereas multimodal association and motor cortices are characterized by a low GLI. The GLI is measured by an automated, computer-controlled scanning procedure, and allows the measurement of a large number of histological sections in a broad range of species. Since the regional heterogeneity of the GLI provides a tool for cortical parcellation, cytoarchitectonic maps of various species included in the three brain collections could be established.^{25–28,43,51–53,55,57,58,61–65,67,68,77–89}

Modeling patterns of divergent brain evolution: comparative allometric approaches

Structural changes along the branches of the tree of life are commonly investigated using allometric approaches exploring the extent to which quantitative changes in brain anatomy (e.g., relative size, white to gray matter ratio, cortical folding) scale disproportionately (nonisometrically or allometrically) between structures.⁹⁰ These studies investigate deviations from a null-model assumption that larger brains are scaled up versions of smaller brains. Phylogenetic comparative methods, mapping extant variation along the branches of a phylogenetic tree,⁹¹ are then used to investigate whether putative allometric trends of particular brain structures are consistent across entire clades or particular to specific branches/species. This approach provides unique insight into quantitative changes in brain anatomy over the course of millions of years of evolution across many species.

Comparative allometric approaches to investigating brain evolution often use the relative size of brain structures as the primary quantitative measure to study evolutionary changes in brain anatomy. This emphasis on the relative size of brain structures stems from a combination of methodological, anatomical, and behavioral reasons. Methodologically, because all primates have the same brain structures, relative sizes of structures provide consistent cross-species measurements. Anatomically, the size of a structure is associated to

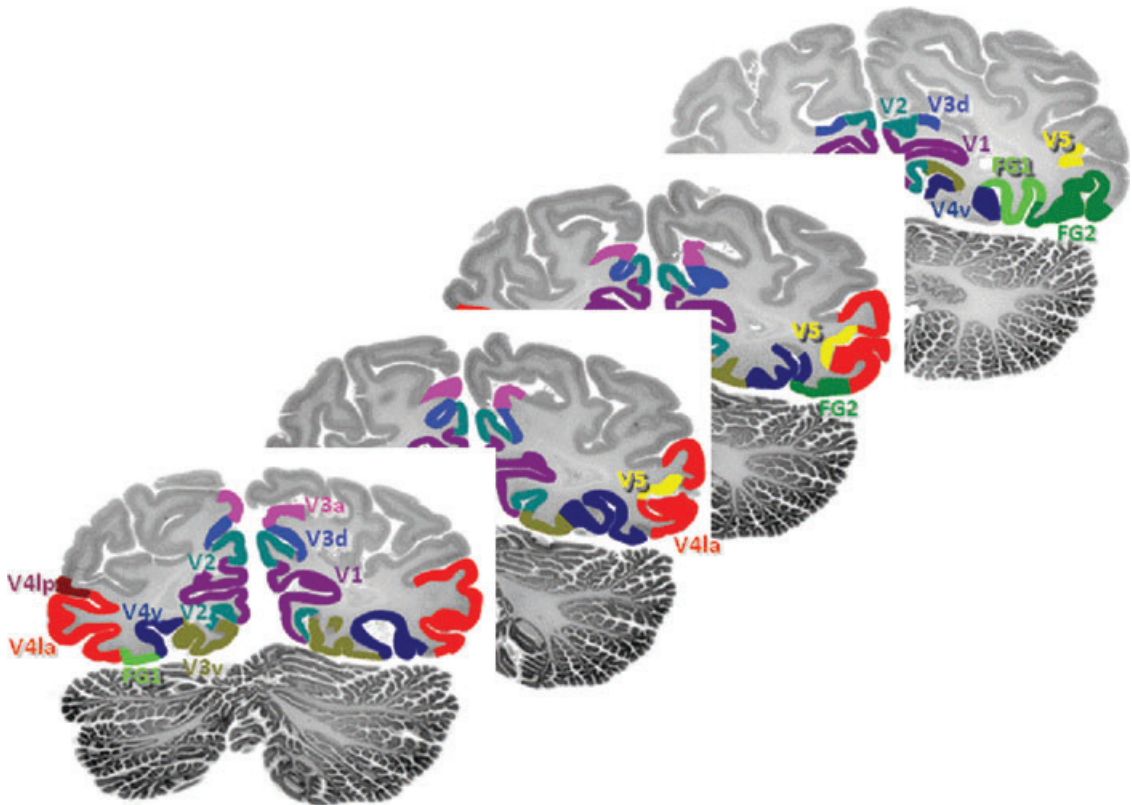


Figure 4. Series of four histological sections in a caudo- (left bottom) to-rostral sequence from a human brain; cytoarchitectonic maps were generated in order to capture intersubject variability in the extent, shape, and localization of cortical areas by comparison with other brains. Cortical areas are labeled by different colors. Please note that the cytoarchitectonic borders of cortical areas are not determined by macroscopical landmarks—for example, funus of a sulcus or crown of a gyrus. Maps have been published by Amunts *et al.*,⁶³ Rottschy *et al.*,⁷⁸ Malikovic *et al.*,⁷⁹ Kujovic *et al.*,⁸⁰ and Caspers *et al.*⁷⁷ V1 = primary visual cortex; V2 = secondary visual cortex; V3a, V3d, V4la, V4lp, V4v, and V5 = extrastriate visual areas; FG1 and FG2 = higher visual areas in the fusiform gyrus.

cytoarchitectonic criteria such as cell morphology, neural/glial ratio, lamination, and myelination.^{92,93} Behaviorally, an animal's brain organization has been suggested to reflect its adaptation to the behavioral requirements of its habitat.⁹⁴ Brain organization is then proposed to reflect an efficient commitment to those neural circuits supporting behavior that increases an individuals' fitness.⁹⁵ Others, however, suggest that conserved developmental timing acts as a constraint on adaptive evolution of individual brain areas and that concerted scaling explains brain structure differences^{96,97} (predicting proportional structural allometry in relation to developmental timing). Empirically, the behavioral adaptation hypothesis has received the most

support, with studies across the animal kingdom (e.g., fish,⁹⁸ bees,⁹⁹ birds,^{100,101} primates,^{102,103} and increasing evidence in humans^{104–107}) indicating a correlation between quantitative changes in brain anatomy and behavioral capacity. Following the empirically most-supported hypothesis by assuming structural/functional correlations and the evolutionary principle that extant variation is the necessary result of past selective pressures, changes in brain structures between species can also be considered as observable extant measures containing essential information on past (and current) behavioral selective pressures,¹⁰⁸ thus providing information on the evolution of behaviors often difficult to observe.¹⁰⁹ In short, relative sizes of brain

structures provide a consistent comparative platform and are generally considered to bear both structural and functional information.

The landmark publication of brain structure volumes in primates⁵ (the brain collection underlying this study is the above described Stephan collection) was the foundation for insights into the evolution of primate brain organization (the composite of relative sizes of its subcomponents) that now dominate fields concerned with the evolutionary neural underpinnings of primate cognitive capacities.^{94–96,110} This single publication on brain structure volumes across primates was the direct catalyst for the emergence of theories that have helped define the nature of “primate intelligence.”^{95,103,111} Based on the original publication of these data, comparative anthropologists and neuroanatomists have demonstrated that evolutionary correlations between relative sizes of brain structures reflect anatomical connectivity,^{95,112} and that primate brain evolution is largely characterized by a disproportionate increase in relative neocortex size.⁹⁵ Neocortex hyperscaling is now commonly regarded as the primary neural substrate underlying “primate intelligence”¹⁰² and as the defining element when inferring primate behavioral selective pressures.¹⁰³

Recent advances in the evolution of primate brain organization

Arguably one of the biggest limitations of allometric analyses of brain structure evolution in primates is that cross-species neocortical and cerebellar data are largely limited to overall size. In primates, the neocortex not only is quite large (70–80% of total brain volume in apes and 60–75% in monkeys), but also is involved in just about any behavior, causing evolutionary analyses of overall neocortical volume to bear little functional information. Moreover, phylogenetic comparative methods that are currently commonplace in cross-species allometric studies infer limited information on species-specific adaptations and almost no information on changes in ancestral branches further down the tree. By focusing on allometric trends across entire clades, these methods miss out on valuable adaptive information that can be inferred from branch-specific rates of evolution.

Recent work has significantly increased the resolution of comparative allometric brain studies

in primates by collecting new data on more detailed and functionally specific brain structures and applying new phylogenetic comparative methods that allow investigating branch-specific rates of evolution. Using the same individuals underlying the landmark publication of brain structure volumes,⁵ Smaers *et al.*¹¹³ parcellated the frontal neocortical lobe from the rest of the neocortex and demonstrated scaling coefficients of 1.19 and 0.99 compared to rest of brain volume, suggesting that neocortex hyperscaling is almost entirely due to frontal lobe hyperscaling. Smaers *et al.*¹⁰⁹ further explored the details of frontal lobe hyperscaling by investigating the scaling coefficients of the prefrontal and frontal motor areas. Results indicated a marked lateralization in prefrontal scaling causing left prefrontal to scale with a coefficient ≈ 1.35 to rest of brain volume and the right prefrontal with a coefficient not significantly different from isometry (≈ 1.05). Frontal motor areas were not indicated to be lateralized or to have a scaling coefficient significantly different from isometry (≈ 1.03 and ≈ 1.09 for left and right hemispheres). These results further define primate “neocortex” hyperscaling in terms of left prefrontal hyperscaling. Prefrontal analyses also suggested apes have evolved a distinct increase in relative white to gray matter volume in left (but not right) prefrontal areas compared to monkeys, and that humans follow this ape trend but to a degree significantly deviating from primate allometry. Together, these results suggest that part of what may separate apes (and humans especially) from other primates is increased left prefrontal white to gray matter ratio possibly related to increased prefrontal structural connectivity and/or increased prefrontal processing speed due to increased myelination of axonal fibre tracts. This structural lateralization in apes may reflect a functional lateralization underlying a cognitive grade shift between apes and other primates.

Furthermore, the application of novel phylogenetic comparative methods has lead to new insights into the evolution of the primate cortico-cerebellar system. Smaers and Vinicius¹¹⁴ recently developed a novel phylogenetic comparative method to infer branch-specific evolutionary rates of phenotypes using a model of evolution that is generally considered to be more suited to model selection (Adaptive Peak model of evolution) than the traditionally used Brownian Motion model.¹¹⁵ This new

approach allows modeling detailed patterns of phenotype evolution providing information on species-specific adaptive patterns. Smaers *et al.*¹¹⁶ used this approach to model the evolution of the cortico-cerebellar system by collecting new data on volumes of the four major subdivisions of the cerebellum (cerebrocerebellum, spinocerebellum, flocculonodular lobe, and cerebellar white matter) and inferring branch-specific evolutionary patterns underlying clade-general relative size correlations between cerebellar and neocortical substructures. Results demonstrate that the ape lineage yields the strongest and most consistent fronto-cerebellar expansion, which provides empirical support for suggestions that “primate intelligence” may not be as unitary as suggested by the traditional exclusive focus on relative neocortex size expansion¹¹⁷ and may involve differential expansion in different clades of different functionally specialized neural systems involving both neocortical and nonneocortical structures.

Conclusion

Recent research combining the collection of new data and the application of new comparative methods has increased the resolution at which we can investigate the divergent evolution of primate brain organization. These studies provide a more detailed picture of quantitative anatomical changes underlying brain evolution across primates and between particular clades and/or species. Investigations of prefrontal¹⁰⁹ and fronto-cerebellar¹¹⁶ evolution have, for example, recently demonstrated neural adaptive shifts between ape and monkey radiation, providing a more detailed picture on which neural substrates may underlie the distinction between cognitive capacities distinguishing apes from other primates. It is clear that without the availability of brain data across many different species, such as those present in the Stephan and Zilles collections, such research is not possible and crucial information on species- and clade-specific patterns of brain evolution cannot be revealed. These studies highlight the fundamental importance of continued efforts in collecting and curating sufficient cross-species data and further suggest that recent advances in comparative methodology together with the exponential advances in primate genotype research provide a unique platform to extract essential medical, neuroanatomical, evolutionary biological, and an-

thropological information from cross-species brain data.

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Supporting information

Additional Supporting Information may be found in the online version of this article:

Table S1. Stephan collection

Table S2. Zilles collection vertebrates

Table S3. Zilles-Amunts collection *Homo sapiens*

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Conflicts of interest

The authors declare no conflicts of interest.

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